

Extra Homework 10

Let $f: \mathbb{R}^m \rightarrow \mathbb{R}$ be a convex function having a minimum $x^* \in \mathbb{R}^m$ and consider the gradient descent method

$$x_{k+1} = x_k - \Delta \nabla f(x_k).$$

Assume that there exist $0 < m < M$ such that for any $x \in \mathbb{R}^m$ the Hessian matrix $H(x)$ of f has all its eigenvalues $\lambda \in [m, M]$.

1. Using the Taylor expansion, prove that

$$f(x_{k+1}) \leq f(x_k) - \Delta \|\nabla f(x_k)\|^2 + \frac{M\Delta^2}{2} \|\nabla f(x_k)\|^2.$$

$$f(x_k + h) = f(x_k) + \nabla f(x_k) \cdot h + \frac{1}{2} h^T H(\xi) h$$

(ξ is a point between x_k and x_{k+1})

$$h = x_{k+1} - x_k \Rightarrow h = x_k - \Delta \nabla f(x_k) - x_k$$

$$h = -\Delta \nabla f(x_k)$$

$$\nabla f(x_k) \cdot h = \nabla f(x_k) \cdot (-\Delta \nabla f(x_k)) = -\Delta \|\nabla f(x_k)\|^2$$

$$\text{The eigenvalues of } H \text{ are } \leq M \Rightarrow h^T H h \leq M \|h\|^2$$

$$\Rightarrow \frac{1}{2} (-\Delta \nabla f(x_k))^T H (-\Delta \nabla f(x_k)) \leq \frac{1}{2} M \Delta^2 \|\nabla f(x_k)\|^2$$

Hessian eigenvalues $\leq M \Rightarrow$

$$\Rightarrow f(x_k + h) \leq f(x_k) + \nabla f(x_k) h + \frac{1}{2} h^T H(\xi) h$$

$$\Rightarrow f(x_{k+1}) \leq f(x_k) - \Delta \|\nabla f(x_k)\|^2 + \frac{M\Delta^2}{2} \|\nabla f(x_k)\|^2$$

2. Choosing the step $\Delta > 0$ in order to minimize the right-hand side in (1), prove that:

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2M} \|\nabla f(x_k)\|^2.$$

Take the right-hand side as a function

$$g(\Delta) = f(x_k) - \Delta \|\nabla f(x_k)\|^2 + \frac{M\Delta^2}{2} \|\nabla f(x_k)\|^2$$

$$g'(\Delta) = [f(x_k)]' - (\Delta \|\nabla f(x_k)\|^2)' + \left[\frac{M\Delta^2}{2} \|\nabla f(x_k)\|^2 \right]' =$$

$$= 0 - \|\nabla f(x_k)\|^2 + \frac{1}{2} M \cdot 2\Delta \|\nabla f(x_k)\|^2 =$$

$$= -\|\nabla f(x_k)\|^2 + M\Delta \|\nabla f(x_k)\|^2$$

$$g'(\Delta) = 0$$

$$-\|\nabla f(x_k)\|^2 + M\Delta \|\nabla f(x_k)\|^2 = 0$$

$$\|\nabla f(x_k)\|^2 (-1 + M\Delta) = 0$$

$$-1 + M\Delta = 0$$

$$M\Delta = 1$$

$$\Delta = \frac{1}{M}$$

Substitute
 Δ back in
formula
from (1)

$$\Rightarrow f(x_{k+1}) \leq f(x_k) - \frac{1}{M} \|\nabla f(x_k)\|^2 + \frac{M \cdot \frac{1}{M^2}}{2} \|\nabla f(x_k)\|^2$$

$$f(x_{k+1}) \leq f(x_k) - \frac{2}{M} \|\nabla f(x_k)\|^2 + \frac{1}{2M} \|\nabla f(x_k)\|^2$$

$$f(x_{k+1}) \leq f(x_k) - \frac{2\|\nabla f(x_k)\|^2 - \|\nabla f(x_k)\|^2}{2M}$$

$$f(x_{k+1}) \leq f(x_k) - \frac{\|\nabla f(x_k)\|^2}{2M}$$

$$f(x_{k+1}) \leq f(x_k) - \frac{1}{2M} \|\nabla f(x_k)\|^2$$

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Using the Taylor Expansion, prove that

$$f(x^*) \geq f(x_k) - \frac{1}{2m} \|\nabla f(x_k)\|^2.$$

$$x^* = x_k + h$$

$$h = x^* - x_k$$

$$f(x^*) = f(x_k) + \nabla f(x_k) \cdot (x^* - x_k) + \frac{1}{2} (x^* - x_k)^T H(f) (x^* - x_k)$$

$$\text{The eigenvalues of } H \geq m \Rightarrow \frac{1}{2} (x^* - x_k)^T H(f) (x^* - x_k) \geq \frac{m}{2} \|x^* - x_k\|^2$$

$$\Rightarrow f(x^*) \geq f(x_k) + \nabla f(x_k) (x^* - x_k) + \frac{m}{2} \|x^* - x_k\|^2 \quad (*)$$

Take the right-hand side as a function

$$g(v) = f(x_k) + \nabla f(x_k)^T \cdot v + \frac{m}{2} \|v\|^2, \text{ where } v = x^* - x_k$$

$$\begin{aligned} g'(v) &= [f(x_k)]' + [\nabla f(x_k)^T \cdot v]' + \left[\frac{m}{2} \|v\|^2 \right]' = \\ &= 0 + \nabla f(x_k) + \frac{m}{2} \cdot 2 \cdot v = \\ &= \nabla f(x_k) + m v \end{aligned}$$

$\|v\|^2 = v \cdot v = v^2$

$$g'(v) = 0$$

$$\nabla f(x_k) + m v = 0$$

$$m v = -\nabla f(x_k)$$

$$v = -\frac{1}{m} \nabla f(x_k)$$

$$\text{So } x^* - x_k = -\frac{1}{m} \nabla f(x_k)$$

Substitute back in formula (*)

$$f(x^*) \geq f(x_k) + \|\nabla f(x_k)\| \left(-\frac{1}{m}\right) \|\nabla f(x_k)\| + \frac{m}{2} \left(-\frac{1}{m}\right)^2 \|\nabla f(x_k)\|^2$$

$$f(x^*) \geq f(x_k) + \|\nabla f(x_k)\| \left(-\frac{1}{m}\right) \|\nabla f(x_k)\| + \frac{1}{2m} \cdot \frac{1}{m^2} \|\nabla f(x_k)\|^2$$

$$f(x^*) \geq f(x_k) - \frac{1}{m} \|\nabla f(x_k)\|^2 + \frac{1}{2m} \|\nabla f(x_k)\|^2$$

$$f(x^*) \geq f(x_k) - \frac{2-1}{2m} \|\nabla f(x_k)\|^2$$

$$f(x^*) \geq f(x_k) - \frac{1}{2m} \|\nabla f(x_k)\|^2$$

4. Combining (2) and (3), prove that

$$f(x_{k+1}) - f(x^*) \leq \left(1 - \frac{m}{M}\right) (f(x_k) - f(x^*)),$$

and hence that $f(x_k) \rightarrow f(x^*)$.

$$\text{Formula (2): } f(x_{k+1}) \leq f(x_k) - \frac{1}{2M} \|\nabla f(x_k)\|^2 \Rightarrow$$

$$\Rightarrow f(x_{k+1}) - f(x_k) \leq -\frac{1}{2M} \|\nabla f(x_k)\|^2 \quad / \cdot (-1)$$

$$f(x_k) - f(x_{k+1}) \geq \frac{1}{2M} \|\nabla f(x_k)\|^2$$

$$\text{Formula (3): } f(x^*) \geq f(x_k) - \frac{1}{2m} \|\nabla f(x_k)\|^2 \Rightarrow$$

$$\Rightarrow f(x^*) - f(x_k) \geq -\frac{1}{2m} \|\nabla f(x_k)\|^2 \quad / \cdot (-1)$$

$$f(x_k) - f(x^*) \leq \frac{1}{2m} \|\nabla f(x_k)\|^2$$

$$\Rightarrow [f(x_k) - f(x^*)] \cdot 2m \leq \underbrace{\|\nabla f(x_k)\|^2}_{\text{Substitute in (2)}}$$

Substitute in (2)

$$f(x_k) - f(x_{k+1}) \geq \frac{1}{2M} \cdot 2m [f(x_k) - f(x^*)]$$

$$f(x_k) - f(x_{k+1}) \geq \frac{m}{M} [f(x_k) - f(x^*)] \quad / \cdot (-1)$$

~~$$f(x_k) - f(x_{k+1}) \geq \frac{m}{M}$$~~

$$-f(x_k) + f(x_{k+1}) \leq -\frac{m}{M} [f(x_k) - f(x^*)] \quad / + (f(x_k) - f(x^*))$$

~~$$-f(x_k) + f(x_{k+1}) + f(x_k) - f(x^*) \leq [f(x_k) - f(x^*)] \left(-\frac{m}{M} + 1\right)$$~~

$$f(x_{k+1}) - f(x^*) \leq \left(1 - \frac{m}{M}\right) (f(x_k) - f(x^*))$$

$$0 < m < M \Rightarrow 0 < 1 - \frac{m}{M} < 1$$

$\Rightarrow$ Each new difference between the value of the function is always a fraction of the previous difference. As k grows bigger, the difference goes to 0, so $f(x_k) \rightarrow f(x^*)$.